A reusable RAG pipeline
A BioASQ reproduction kit
A model-organism case study

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Top-ranked retrieval in BioASQ 14b Phase A

Ranked 1st in 3 batches · preliminary results

01 A reusable RAG pipeline

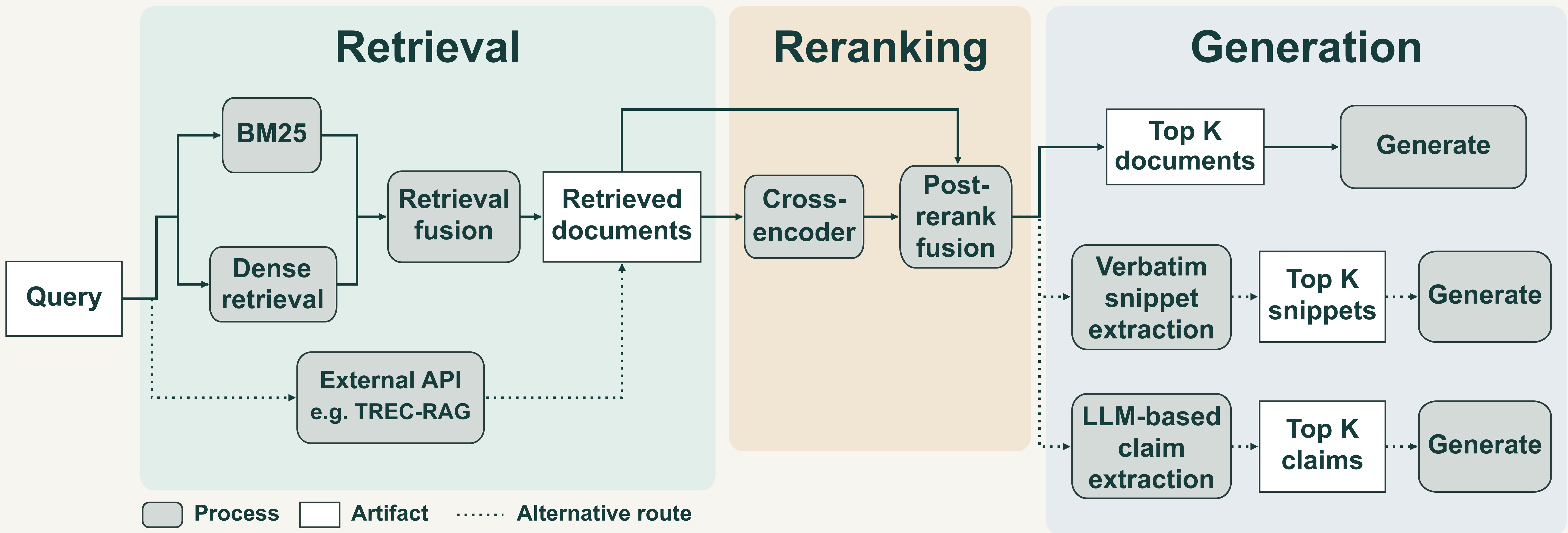
github.com/fulaibaowang/RAG-scripts

Configurable
Pipeline stages and settings in one config

Reproducible
Docker image, pinned dependencies

Portable
BioASQ, TREC-RAG, your corpus

Agent-friendly
Built-in guidance for AI coding assistants



02 A BioASQ reproduction kit

github.com/fulaibaowang/BioASQ

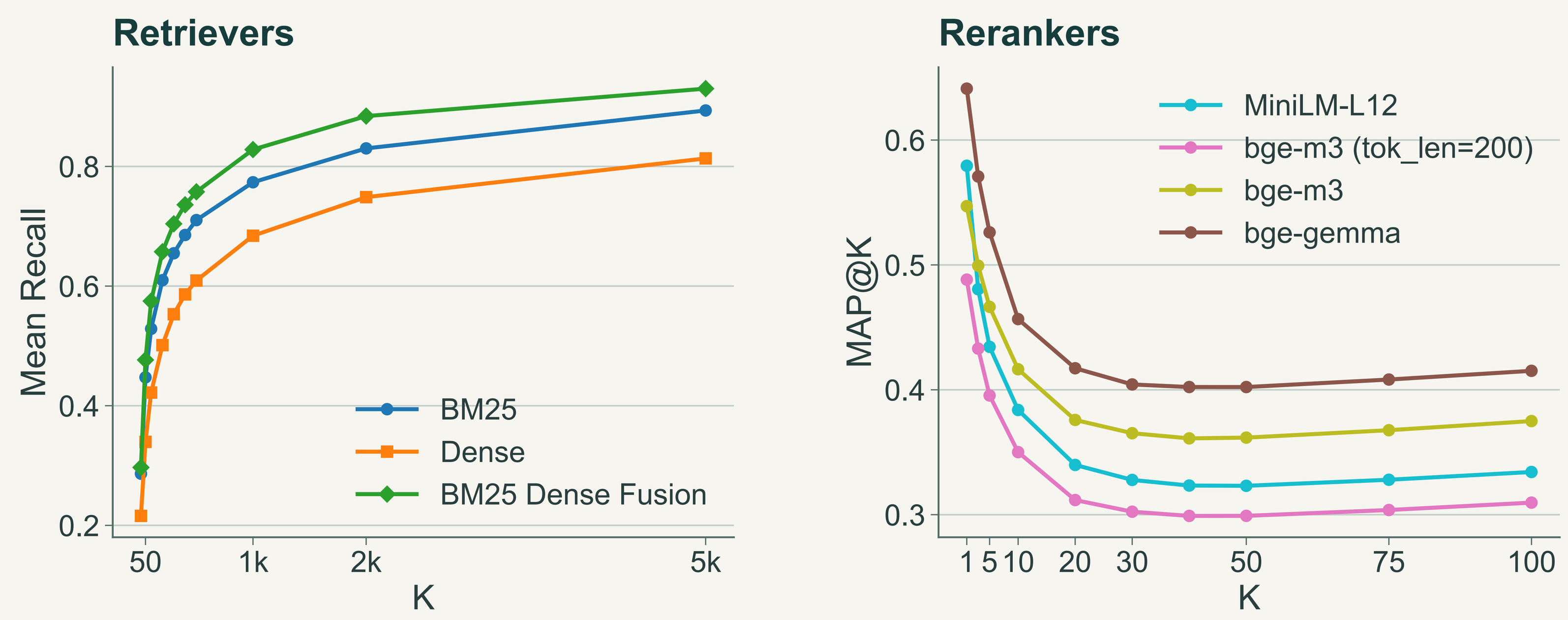
Scripts and configurations to reproduce our BioASQ retrieval runs.

Build index → Run pipeline → Export JSON

One batch of 80 questions · one A100 GPU

16	~80 GB	~1.5 h	~150 GB
CPU cores	RAM	retrieve + rerank	one-time: corpus, indexes, container

Diagnostic plots included · detailed analysis in the working note



Examples on BioASQ 13b, batches 1–4 averaged

03 A model-organism case study

github.com/fulaibaowang/dictycite

Why is our system called DictyCite?

Dicty (*Dictyostelium*) is a model organism in biology. We built a literature retrieval benchmark and case study for it from curated biological annotations.

> Our *Dictyostelium* case study suggests that tailoring literature retrieval to a biological domain can be worthwhile, using curated annotations to enrich queries and choosing retrievers and rerankers that fit the task.

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UNIVERSITY OF LJUBLJANA · FRI Faculty of Computer and Information Science

SMASH machine learning for science and humanities postdoctoral program